

Maths

Q1) $\sqrt{3}, 2\sqrt{3}, 3\sqrt{3} \dots$

$$a = \sqrt{3}$$

$$d = \sqrt{3}$$

$$T_n = a + (n-1)d$$

$$T_{100} = \sqrt{3} + (100-1)d$$

$$T_{100} = \sqrt{3} + 99\sqrt{3}$$

$$= 100\sqrt{3}$$

Q2)

→ infinite

Q3)

→ n^{th} term = $8n - 5$

$$a_n = 8(1) - 5$$

$$= 8 - 5$$

$$= 3$$

$$a_2 = 8(2) - 5$$

$$= 16 - 5$$

$$= 11$$

~~Q4th term =~~

28th term =

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

$$= \frac{28}{2} (2(3) + (28-1)8)$$

$$= 14 (6 + 27(8))$$

$$= 14 (6 + 216)$$

$$= 14(222) = 3108$$

Q4)

→ $OP = OO$ --- Radii of same circle

ΔPOO --- Isosceles triangle

$$\angle OPO = \angle OOP \quad \text{--- (i)}$$

$$\text{Let } \angle OPO = \angle OOP = x \quad \text{--- (ii)}$$

In ΔPOO

$$\angle OPO + \angle OOP + \angle POO = 180^\circ$$

$$x + x + 70 = 180$$

$$2x = 180 - 70$$

$$2x = 110$$

$$x = \frac{110}{2}$$

$$x = 55^\circ$$

$$\therefore \angle OPT = 90^\circ$$

--- Tangent

$$= \angle OPT = \angle OPO + \angle TPO$$

$$= 90^\circ = 55^\circ + \angle TPO$$

$$= \angle TPO = 90 - 55$$

$$= \angle TPO = 35^\circ$$

05) 3, 15, 27, 39, ... $a=3, d=12$
 $n=54$

$$a_n = a + (n-1)d$$

$$\begin{aligned} a_{54} &= a + (n-1)d \\ &= 3 + (54-1)12 \\ &= 3 + 53 \times 12 \\ &= 639 \end{aligned}$$

Term 132 is more than its
 54^{th} term

$$\text{So } 639 + 132 = 771$$

$$a_n = 771$$

$$771 = a + (n-1)d$$

$$771 = 3 + (n-1)12$$

$$771 - 3 = (n-1)12$$

$$768 = (n-1)12$$

$$(n-1) = \frac{768}{12}$$

$$(n-1) = 64$$

$$\boxed{n = 65}$$

0.6) given = PR = 12

$$op = 5$$

$$oo = 4$$

$$\text{find} = RO$$

$$op = oo + top$$

$$op = 4 + 5$$

$$op = 9$$

$$\begin{aligned}
 RO^2 &= PR^2 + OR^2 \\
 RO^2 &= 12^2 + 9^2 \\
 RO^2 &= 144 + 81 \\
 RO^2 &= 225 \\
 RO &= 15 \\
 \boxed{RO = 15}
 \end{aligned}$$

$$\begin{aligned}
 RO^2 &= RO^2 - OR^2 \\
 RO^2 &= 15^2 - 4^2 \\
 RO^2 &= 225 - 16 \\
 RO &= \sqrt{209}
 \end{aligned}$$

Q.7) $AP = 174 + 261 - 957$

$$a = 174 + (10-1) \cdot 87 = 122$$

$$d = 87$$

$$an = 957$$

$$an = a + (n-1)d$$

$$957 = 174 + (n-1)(87)$$

$$957 - 174 = (n-1)(87)$$

$$783 = (n-1)(87)$$

$$(n-1) = \frac{783}{87}$$

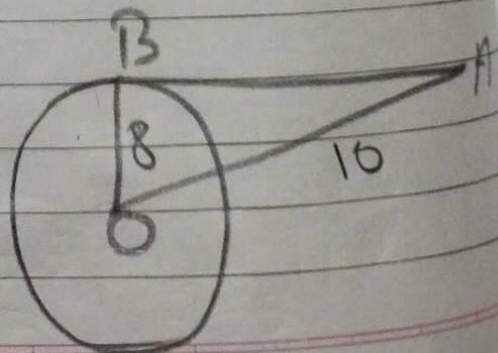
$$(n-1) = 9$$

$$\boxed{n = 10}$$

Q.8) $AO = 10$

$$BO = 8$$

$$AB^2 = AO^2 - BO^2$$



$$AB^2 = 10^2 - 8^2$$

$$AB^2 = 100 - 64$$

$$AB^2 = 36$$

$$AB = 6 \text{ cm}$$

Q 9)

$$a_2 = 14$$

$$a_3 = 18$$

$$d = 4$$

$$a_2 = a + d = 14$$

$$a + 4 = 14$$

$$a = 10$$

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

$$S_{51} = \frac{51}{2} (2(10) + (51-1) \cdot 4)$$

$$\frac{51}{2} (20 + 200)$$

$$51 \times 220$$

$$51 \times 110$$

$$S_{51} = 5610$$